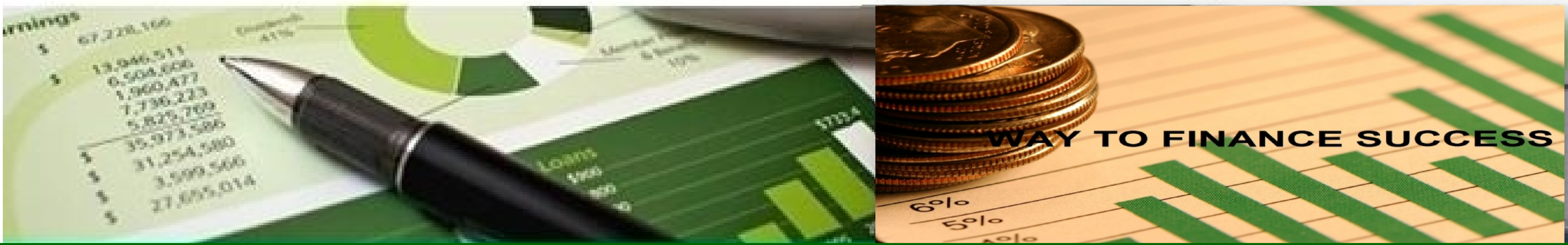
Way To Finance Success

MIND MAP CFA® EXAM PRE

LEVEL 1

2016



MIND MAPS

For learning CFA® Exam

- Full Mind Maps of 10 Topics
- Brief but Full
- Easy to Memorize
- Effective to Revise

Website: http://waytofinancesuccess.com
Email: waytofinancesuccess@gmail.com

1. Code Of Ethics And Standards Of Professional Conduct

- a.
  - All CFA Institute members and candidates are required to comply with the Code and Standards
  - Structure of the CFA Institute Professional Conduct Program
    - Basic structure for enforcing the Code and Standards
      - The CFA Institute Bylaws
      - Rules of Procedure
        - Based on two primary principles
          - Fair process to member and candidate
          - Confidentiality of proceedings
    - Professional Conduct program (PCP)
      - The CFA Institute Board of Governors
        - Maintains oversight and responsibility
        - Through the Disciplinary Review Committee (DRC)
          - Is responsible for the enforcement of the Code and Standards
      - The CFA Designated Officer
        - Directs professional conduct staff
          - Conducts professional conduct inquiries
    - An inquiry can be prompted by several circumstances
      - Selfdisclosure
      - Written complaints
      - Evidence of misconduct
      - Report by a CFA exam proctor
      - Analysis of exam materials and monitoring of social media by CFA Insitute
  - Process for the enforcement of the Code and Standards
    - When an inquiry is initiated
      - The Professional Conduct staff conducts an investigation that may include
        - Requesting a written explanation from the member or candidate
        - Interviewing
          - The member or candidate
          - Complaining parties
          - Third parties
        - Collecting documents and records in support of its investigation
      - Upon reviewing the material obtained during the investigation, the Designated Officer may
        - Conclude the inquiry with no disciplinary sanction
        - Issue a cautionary letter
        - Continue proceedings to discipline the member or candidate
          - If finding that a violation of the Code and Standards occurred, the Designated Officer proposes a disciplinary sanction
            - Accepted by member
            - Rejected by member
              - The matter is referred to a hearing by a panel of CFA Institute members
          - If sanction is imposed
            - condemnation by the member's peers
            - suspension of candidate's continued participant in the CFA program
- b,c.
  - Six components of the Code of Ethics
    - Act with integrity, competence, diligence, respect and in an ethical manner
    - Integrity of investment profession & interest of clients above personal interest
    - Care & judgment
    - Practice ethics & encourage others to practice
    - Integrity & viability of the global capital markets
    - Professional competence
  - Seven Standards of Professional Conduct
    - Professionalism
    - Integrity of Capital markets
    - Duties of Clients
    - Duties to Employers
    - Investment analysis, Recommendations & Actions
    - Conflict of interest
    - Responsibilities as a CFA Institute member or CFA Candidate

2.1 Standard I PROFESSIONALISM

- **A. Knowledge of the law**
  - Guidance
    - Understand and comply with applicable laws and regulations
    - Code and Standards vs. Local law
      - Follow stricter law and regulation
    - Participation or association with violations by others
      - Responsible for violations in which they knowingly participate or assist
      - Dissociate from illegal, unethical activities
        - Leave employers (in extreme case)
      - Intermediate steps
        - Attempt to stop the behavior by bringing it to the attention of employer through a supervisor or compliance department
        - May consider directly confronting the involved individuals
        - If not successful,--> step away and dissociate from the activity by
          - Removing their name from written reports
          - Asking for a different assignment
      - Inaction with continued association may be construed as knowing participation
      - Not required reporting violations to government, CFAI, but advisable in some cases or required by laws in others
  - Recommended procedures for compliance (RPC)
    - Members and candidates
      - Stay informed
      - Review procedures
      - Maintain current files
      - When in doubt, seek advice of compliance personnel or legal counsel
      - When dissociating from violations, --> Document any violations and urge firms to stop them
    - Firms
      - Develop and/or adopt a code of ethics
      - Make available to employees info that highlights applicable laws and regulations
      - Establish written procedures for reporting suspected violation of laws, regulations or company policies
  - Application
- **B. Independence and objectivity**
  - Guidance
    - Maintain independence and objectivity in professional activities
    - How to cope with external and internal pressures
      - External pressures
        - By benefits
          - Gifts, Invitations to lavish functions, Tickets, Favors, Job referrals, Allocation of shares in oversubscribed IPOs...
        - From public companies
          - To issue favorable reports
        - From Buyside clients
          - May try to pressure sellside analysts
      - Internal pressures
        - From their own firms
          - e.g. to issue favorable research reports/ recommendations for certain companies
        - Investmentbanking relationships
          - to issue favorable research on current or prospective investmentbanking clients
          - Conflicts of interest
      - -->
        - Modest gifts and entertainment are acceptable but special care must be taken
          - must disclose to employers
        - Best practice: reject any offer of gifts, threatening independence and objectivity
        - Recommendations must
          - convey true opinions
          - free of bias from pressures
          - be stated in clear and unambiguous language
        - Portfolio managers must respect and foster honesty of sellside research
    - Issuerpaid research
      - Is fraught with conflicts
      - Analysts
        - Must engage in thorough, independent, and unbiased analysis
        - Must fully disclose potential conflicts, including the nature of compensation
        - Must strictly limit the type of compensation they accept for conducting research
        - Best practice
          - Accept only flat fee for their work prior to writing the report
          - Without regard to conclusions or recommendations
  - RPC
    - Protect integrity of opinions
    - Create a restricted list
    - Restrict special cost arrangements
    - Limit gifts
    - Restrict employee investments
      - Equity IPOs
      - Private placements
    - Review procedures
    - Written policies on independence and objectivity of research
- **C. Misrepresentation**
  - Guidance
    - Definition of "Misrepresentation"
      - any untrue statement or omission of a fact
      - or any false or misleading statement
    - Must not knowingly make misrepresentation or give false impression in
      - oral representations, advertising
      - electronic communications
      - written materials
    - Must not misrepresent any aspect of practice, including
      - qualifications or credentials, services
      - performance record
      - Without regard to conclusions or recommendations
      - characteristics of an investment
      - any misrepresentation relating to member's professional activities
    - Must not guarantee clients specific return on investments that are inherently volatile
    - Standard I(C) prohibits plagiarism in preparation of material for distribution to employers, associates, clients, prospects, general publish
  - RPC
    - Written list of available services, description of firm's qualification
    - Designate employees to speak on behalf of firm
    - Prepare summary of qualifications and experience, list of services capable of performing
    - To avoid plagiarism
      - Maintain copies
      - Attribute quotations
      - Attribute summaries
- **D. Misconduct**
  - Guidance
    - Address conduct related to professional life
    - Violations
      - Any act involving lying, cheating, stealing, other dishonest conduct that reflects adversely on member's professional activities would be violation
      - Conduct damaging trustworthiness or competence (include behaviour may not be illegal but negatively affect a member to perform responsibility such as abusing alcohol during lunch hours)
      - Abuse of the CFA Institute Professional Conduct Program
      - Involved in personal bankruptcy is not automatically assumed to be in violation but bankruptcy involve fraudulent or deceitful business conduct may be a violation
  - RPC
    - Develop and/or adopt a code of ethics
    - Disseminate to all employee a list of potential violations
    - Check references of potential employees

3+4 GIPS

Note: GIPS standards are printed in their entirety in the readings, but the Level I candidate is required only to know the material through the end of Section II.0 "Fundamental of Compliance."

Introduction to Global Investment Performance Standards (GIPS)

- a1. Why were the GIPS Standards created?
  - The financial markets and investment management industry are becoming increasingly global
- a2. Who can claim compliance?
  - Only investment management firms that actually manage assets
- a3. Who benefit from Compliance?
  - Prospect clients and investment management firms
- b. Construction & purpose of Composites
  - A composite is an aggregation of discretionary portfolios into a single group that represents a particular investment objectives or strategy
  - A composite must include all actual, fee-paying discretionary portfolios managed in accordance with the same investment objective or strategy
  - Composites must include new portfolios on a timely and consistent basis after the portfolio comes under management
  - Firms may set minimum asset levels for inclusion in a portfolio, but changes to a composite-specific minimum asset level are not permitted retroactively.
  - Terminated portfolios must be included in the historical returns of appropriate composites
- c. Verification
  - Increase the level of confidence that a firm claiming GIPS compliance did adhere to GIPS
  - Improve a firm's internal policies and procedures with regard to all aspects of complying with the GIPS standards.
  - Firms are encouraged but not required to undertake the verification process
  - A single verification report is issued for the entire firm. Verification cannot be carried out for a single composite
  - Firms that have been verified are encouraged to add a disclosure to composite presentations or advertisements stating they have been verified: "[name of firm] has been verified for the periods [insert dates] by [name of verifier]. A copy of the verification report is available upon request."

Key features of the GIPS standards & fundamentals of compliance

- GIPS Objectives
  - To obtain global acceptance of calculation and presentation standards in a fair, comparable format with full disclosure
  - To ensure consistence, accurate investment performance data
  - To promote fair competition among investment management firms
  - To promote global "self regulation"
- Key characteristics
  - To claim GIPS, investment management firms must define its "firm"
  - Require Firms to include all actual fee paying, discretionary portfolios in composites defined according to similar strategy / investment objectives
  - Rely on integrity of input data
  - If an investment firm applies GIPS in a performance situation that is not addressed specifically by GIPS/ is open to interpretation, disclosures other than those required by GIPS may be necessary
  - GIPS do not address every aspect of performance measurement, valuation, attribution or cover all asset classes
- Firms must meet full compliance to claim GIPS
  - Compliance cannot be achieved on a single product, portfolio, or composite
- Effective date
  - The effective date of the revised Standards is 1 Jan 2011. Presentations that include performance results for periods after 31 Dec. 2005 must meet all the requirements of the revised GIPS. Performance presentations that include results through 31 Dec. 2005 maybe prepared in compliance with the 1999 version of GIPS.
- Documents policies and procedures
  - Firms must document, in writing, their polices and procedures used in establishing and maintaining compliance with all requirements of GIPS
- Claims of compliance
  - Once a firm has meet all the required requirements of GIPS , use this statement to declare: "[Insert name of firm] has prepared and presented this report in compliance with the Global Investment Performance Standards (GIPS)."
  - If not meet all the requirements, cannot state:"...in compliance with GIPS except for..."
  - Statements referring to the calculation methodology used in a composite presentation as being "in accordance [or compliance] with the Global Investment Performance Standards" are prohibited .
  - Statements referring to the performance of a single, existing client as being "calculated in accordance with the Global Investment Performance Standards" are prohibited except when a GIPS complaint firm reports the performance of an individual account to the existing client
- Firm fundamental responsibilities
  - provide a compliant presentation to all prospect clients, cannot choose to whom they want to present compliant performance
  - provide a complete list and description of all of the firms' composites to any client that makes such a request
    - must list discontinued composites on the firms' list of composites for at least 5 years after discontinuation

Major sections of GIPS standards

- Fundamentals and Compliance
- Input data
  - Consistency of input data is critical to effective compliance with GIPS and establish a foundation for full, fair and comparable performance presentations
- Calculation methodology
  - Uniformity in methods used to calculate returns to achieve comparability among firms
- Composite construction
  - composite return is the asset-weighted average of all the portfolios' performance results
- Disclosures
  - allow firms to elaborate on the raw numbers and give the end user the proper context to understand
  - No "negative assurance" is needed for non-applicable disclosures
- Presentation and reporting
- Real estate
- Private equity
  - Refers to investments in non-public companies that are in various stages of development and venture investing, buyout investing and mezzanie financing
- Wrap Fee/ Separately Managed Account (SMA) portfolios.
  - Wrap fees are a type of bundle fee and are specific to a particular investment product
  - is charged by a wrap fee sponsor for investment management services and included trading expenses that cannot be separately identified
  - can be all-inclusive, asset-based fees and may include a combination of investment management fees, trading expenses, custody fees and/or administration fees
  - A wrap fee portfolio is sometimes referred to as a "separately managed account (SMA) or "managed account"

How are GIPS standards implemented in countries with existing standards for performance reporting

- If local/country specific law or regulation conflicts with GIPS
  - Comply with local law or regulation conflicts with GIPS
  - Make full disclosure of the conflict
  - Note: this differs from Standards of Professional Conduct in which the stricter of local laws or Standards of Professional Conduct prevails

The scope of the GIPS

- Investment firm definition
  - Firms from any country may come into compliance with GIPS
  - GIPS must be applied on the firm-wide basis. Firm must be defined as an investment firm, subsidiary, or division held out to clients as a distinct business entity
  - Total firm assets must be the aggregate of the market value of all discretionary and non-discretionary assets under management. This includes both fee-paying and non-fee-paying assets
- Historical performance record
  - Firms must initially show GIPS compliant history for a minimum of 5 years, or since inception if the firm has been in existence for less than 5 years.
  - After 5-year compliant history has been achieved, firms must add an additional year of performance each year until 10-year performance record is established, at a minimum
  - A firm may link non-GIPS compliant performance to its compliant history as long as
    - only GIPS compliant performance is presented for periods after 1 Jan. 2000; and
    - Firm discloses non-compliance period and explain how it is not in compliance with GIPS
  - Firms previously claiming compliance with an Investment Performance Council-endorsed Country Version of GIPS are granted reciprocity to claim compliance with GIPS for historical periods prior to 1 Jan. 2006

5. TIME VALUE OF MONEY

- **a. Interest rate, considered as**
  - Required rate of return
    - equilibrium interest rate for a particular investment
  - Discount rate
    - for calculating the present value of future cash flows
  - Opportunity cost
- **b. Interest rate**
  - Nominal risk-free rate = real risk-free rate + expected inflation rate
    - real risk-free rate is a theoretical rate on a single-period loan when there is no expectation of inflation.
  - Several risks of securities
    - default risk
      - a borrower will not make the promised payments in timely manner
    - liquidity risk
      - receiving less than fair value if an investment must be sold for cash quickly
    - maturity risk
      - Longer-term bonds have more risk than shorter-term bonds
  - -->The required rate of return on a security = real risk-free rate + expected inflation rate + default risk premium + liquidity premium + maturity risk premium
- **c,d. EAR**
  - represents the annual rate of return actually being earned *after adjustments have been made for different compounding periods*
  - $ERA = (1 + \text{periodic rate})^m - 1$
    - Where:
    - Periodic rate = stated annual rate/m
    - m = the number of compounding periods per year
  - Non-annual time value of money problems
    - divide the stated annual interest rate by the number of compounding periods per year, *m*, and multiply the number of years by the number of compounding periods per year
- **e. CF calculations**
  - Future value
    - $FV = PV(1 + I/Y)^N$
  - Present value
    - $PV = FV/(1 + I/Y)^N$
  - Annuity
    - a series of equal cash flows that occurs at evenly spaced intervals over time.
    - Ordinary Annuity
      - occur at the end of each time period.
    - Annuity Due
      - occur at the beginning of each time period.
        - FV of Annuity Due = FV of Ordinary Annuity x (1+ I/Y)
        - PV of Annuity Due = PV of Ordinary Annuity x (1+ I/Y)
  - PV of a Perpetuity
    - $PV_{perpetuity} = \frac{PMT}{I/Y}$
  - Uneven CF
    - Discount each individual cash flows
    - Use CF function in Calculator
- **f1. Use time line**
  - to solve many types of time value of money problems
  - Loan payment and Amortization
    - Find PMT
    - Find N
    - Find I/Y
    - Amortization table
  - Other applications
    - Rate of compound growth
    - Number of periods for specific growth
    - Funding a future obligation
  - Connection between PV, FV & series of CF
    - the sum of the present values of the cash Rows is the present value of the series. The sum of the future values (at some future time = n) of a series of cash flows is the future value of that series of cash flows.
    - The *cash flow additivity* principle refers to the fact that present value of any stream of cash flows equals the sum of the present values of the cash flows

6. DISCOUNTED CASH FLOW APPLICATIONS

Convert among these yields

- $r_{MM} = \frac{360 \times r_{BD}}{360 - (t \times r_{BD})}$
- $HPY = r_{MM} \times \frac{t}{360}$
- $HPY = (1+EAY)^{t/365} - 1$
- $EAY = (1+HPY)^{365/t} - 1$
- $BEY = [(1 + EAY)^{1/2} - 1] \times 2$

Yields of T-bills

- Bank discount yield
  - Not much meaningful
    - 1. Based on face value, not price
    - 2. Use 360-day
    - 3. Use simple interest, ignore reinvestment of interest
  - $r_{BD} = \frac{D}{F} \times \frac{360}{t}$
  - Where:
    r_{BD} = the annualized yield on a bank discount basis
    D = the dollar discount, which is equal to the difference between the face value of the bill and the purchase price
    F = the face value (par value) of the bill
    t = number of days remaining until maturity
    360 = bank convention of number of days in a year
- Holding period yield
  - $HPY = \frac{P_1 - P_0 + D_1}{P_0} = \frac{P_1 + D_1}{P_0} - 1$
  - Where:
    Po = initial price of the the instrument
    P_1 = price received for instrument at maturity
    D_1 = interest payment (distribution)
- Effective annual yield
  - $EAY = (1 + HPY)^{365/t} - 1$
- Money market yield
  - r_{MM} = HPY x (360/t)
- Bond equivalent yield
  - BEY = 2 x semi annual discount rate

Calculate, Interpret, Decision rule

- NPV
  - the PV of the cash flows less the initial (time = 0) outlay
  - $NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}$
    - where:
      CFt = the expected net cash flow at time *t*
      N = the estimated life of the investment
      r = the discount rare (opportunity cosr of capital)
  - Decision rules
    - Accept projects with a positive NPV
    - Reject projects with a negative NPV
    - Two mutually exclusive projects: accept higher positive NPV
- IRR
  - is the discount rate that make the NPV of a project equal to zero
    - $CF_0 + \frac{CF_1}{(1+IRR)} + \frac{CF_2}{(1+IRR)^2} + ... + \frac{CF_t}{(1+IRR)^t} = 0$
  - Problems
    - Conflict with NPV due to
      - Different project size: the smaller projects may have higher IRR but their contribution to the firm value may be smaller compared to the larger projects
      - Differen timing of cash flows
    - Multiple IRR or No IRR
      - When CFA pattern is unconventional
    - Unrealistic assumptions
      - IRR method: project cash flows are assumed to reinvest at IRR while with NPV it is assumed to reinvest at market rate
      - --> at the bottom lines: use NPV
  - Decision rules
    - Accept projects with an IRR > the firm's (investor's) required rate of return.
    - Reject projects with an IRR < the firm's (investor's) required rate of return.
- For single project, IRR and NPV lead to exactly the same decision

HPR

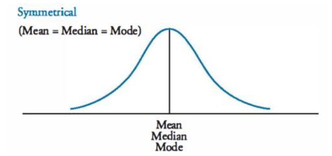
- is the percentage change in an investment over the period of holding

Portfolio rate of return

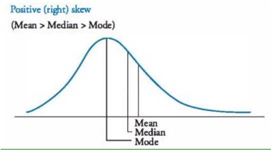
- Money Weighted
  - defined as the IRR
  - More appropriate if manager has complete control over cash in/out
- Time weighted (chain-link)
  - measures compound growth
  - Not affected by cash in/out
  - Preferred method
  - 3 steps
    - Value the investment immediately after any withdrawals or deposits, divide the overall investment horizon into subperiods
    - Calculate HPR for each subpediod
    - Compute the geometric mean

7. Statistical Concepts and Market Returns

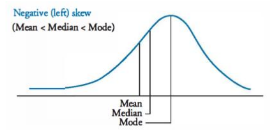
- **a.**
  - Statistical methods
    - Statistics is used to refer to data and to the methods we use to analyze date
    - Descriptive statistics
      - to summarized the important characteristics of large data sets
    - Inferential statistics
      - pertain to the procedures used to make forecasts, estimates, or judgement about a large set of data
  - Population vs. Sample
    - A population is defined as the set of all possible members of a stated group
      - Population parameters
    - A sample is defined as a subset of the populations of interest
      - Sample statistics
      - The most frequently concerned
        - mean (measures of central tendency) which addresses return
        - Var (measures of variation around center) which addresses risk
  - Types of measurement scales
    - Nominal scales
      - Classify or count observations with no particular or ranking
    - Ordinal scales
      - Specified characteristics are used to categorize observations band involve ranking
      - no information on the difference among categories
    - Interval scales
      - Like ordinal scales + the differences between scale values are equal -> scale values can be added and subtracted
      - No true zero point
        - cannot build meaningful ratios
    - Ratio scales
      - Provide ranking, equal differences between scale values and true zero point
- **b.**
  - Parameter vs. Sample statistic
    - A parameter is a measure used to describe a characteristic of a population
    - A sample statistic is used to measure a characteristic of a sample
  - Frequency distribution
    - Definition
      - A tabular presentation of statistical data that aids the analysis of large data sets
    - Construction of a frequency distribution
      - 3 steps
        - 1. Define interval
        - 2. Tally the observations
        - 3. Count the observations and then calculate
- **c.**
  - Absolute frequency
  - Relative frequency
    - calculated by dividing the absolute frequency of each return interval by the total number of observations.
  - Cumulative absolute frequency
    - summing the absolute frequencies starting at the lowest interval and progressing through the highest.
  - Cumulative relative frequency
    - summing the relative frequencies starting at the lowest interval and progressing through the highest.
- **d.**
  - Histogram
    - bar chart
  - Frequency polygon
    - line chart
- **e. Measures of central tendency**
  - Mean
    - Population mean: $\mu = \frac{\sum_{i=1}^{N} X_i}{N}$
    - Sample mean: $\overline{X} = \frac{\sum_{i=1}^{N} X_i}{N}$
    - Arithmetic mean
      - the measure of central tendency for which the sum of the deviations from the mean is zero
      - Sum of mean deviation = $\sum_{i=1}^{n}(X_i - \overline{X}) = 0$
    - Weighted mean (portfolio return): $\overline{X}_W = \sum_{i=1}^{n} w_i X_i = (w_1X_1 + w_2X_2 + ... + w_nX_n)$
    - Geometric mean
      - $G = \sqrt[n]{X_1 x X_2 x \dots X_n}$ (compound growth)
      - $1 + G_r = \sqrt[n]{(1+R_1)x(1+R_2)x \dots x(1+R_n)}$ (return data set)
    - Use of arithmetic or geometric mean when determining investment returns
    - Harmonic mean (cost of shares): $\frac{N}{\sum_{i=1}^{n}\frac{1}{X_i}}$
    - Harmonic < geometric < arithmetic
  - Median
    - value of middle item in a set of sorted items
    - not affected by extreme value but more difficult to find out
  - Mode
    - No mode
    - Unimodal, bimodal, trimodal --> the only measure can be used with nominal scale
    - Model interval --> for continuous distribution
- **f. Quantile**
  - value at or below which a portion of the data distribution lies
  - Quartiles
    - into quarters
  - Quintile
    - into fifths
  - Decile
    - into tenths
  - Percentile (100)
    - Ly =(n+1) $\times$y/100
- **l. Kurtosis**
  - Calculate
    - Sample kurtosis = $\frac{1}{n}\frac{\sum_{i=1}^{n}(X_i - \overline{X})^4}{s^4}$
    - Excess kurtosis = sample kurtosis - 3
  - Compared with normal distribution
    - Leptokurtic: more peaked, fatter tails (excess kurtosis > 0) --> more risk
    - Platykurtic: less peaked (excess kurtosis < 0)
    - Mesokurtic: identical (excess kurtosis = 0)
- **j,k. Shape of distribution**
  - Symmetrical



    - mean=median=mode
    - the frequency of experiencing losses and gains are the same
  - Nonsymmetrical (Skewness) (because of outliers)
    - Types
      - $S_k = \frac{1}{n}\frac{\sum_{i=1}^{n}(X_i - \overline{X})^3}{s^3}$



      - Positively skewed (Sk>0)



      - Negatively skewed (Sk<0) --> more risk
- **i. Relative dispersion**
  - CV (Coefficient of Variation)
    - $CV = \frac{s_x}{\overline{X}} = \frac{\text{standard deviation of x}}{\text{average value of x}}$
  - Sharpe Ratio / Reward-to-Variability ratio
    - Sharpe ratio = $\frac{\bar{r}_p - r_f}{\sigma_p}$
    - Limitations
      - Negative Sharpe ratio
      - Not suitable with asymmetric return distribution
- **h. Chebyshev's inequality**
  - For any distribution with finite variance, the percentage of observations lie within k standard deviation of the mean is at least 1-1/(k^2)
  - 36%: +/-1.25k
  - 56%: +/-1.50k
  - 75%: +/-2k
  - 89%: +/-3k
  - 94%: +/-4k
- **g. Dispersion (measure of risk)**
  - Range = Max - Min
    - Easy to compute
    - affected by extreme value
    - no info on how data is distributed
  - $MAD = \frac{\sum_{i=1}^{n}|X_i - \overline{X}|}{n}$
    - better than range
    - less sophisticated than Var and Sd
  - Variance & Standard deviation
    - Population: $\delta^2 = \frac{\sum_{i=1}^{n}(X_i - \mu)^2}{N}$
    - Sample: $s^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$
  - Semivariance and semideviation
    - $\text{Semivariance} = \frac{\sum_{X_i \le \overline{X}}(X_i - \overline{X})^2}{n-1}$

22. FSA Introduction

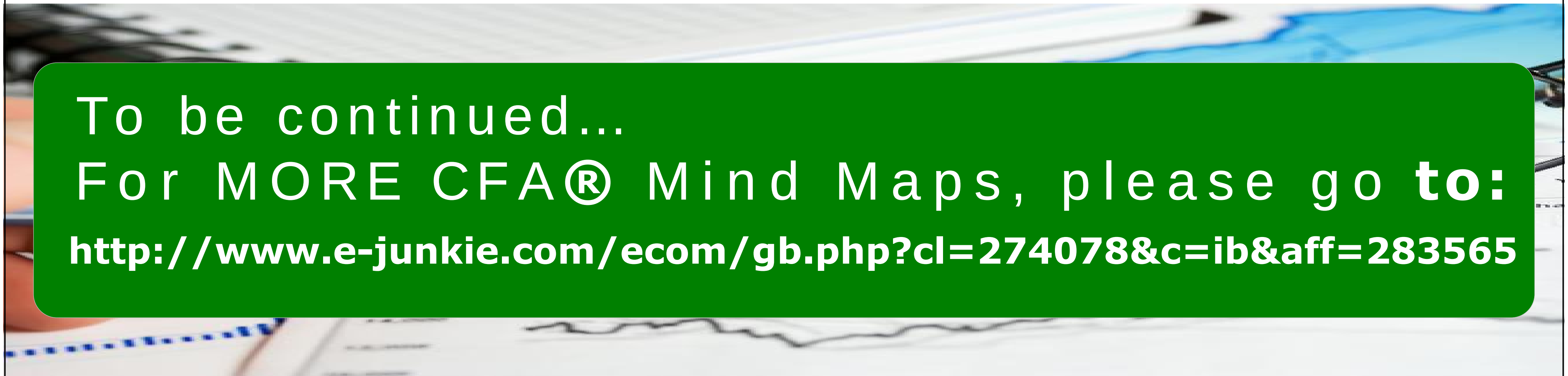
- **Roles of FR & FSA**
  - FR
    - Element
      - Financial Statement
      - Additional disclosures required by regulatory
      - Any commentary by management
    - Role of FR
      - Financial position
      - Firm's performance
      - Changes in financial position
      - Useful to a wide range of users in making economic decisions
  - Roles of FSA
    - Use info in a company's Fin Statements
    - Use other relevant info
    - > To evaluate past, current, and prospective performance & fin position
      > To make economic decisions
- **Role of key FS**
  - Income Statement
    - Revenues
    - Expenses
    - Gains and Losses
  - Balance Sheet (A=L+OE)
    - Assets
    - Liabilities
    - Owners' equity
  - CF statement
    - CFO
    - CFI
    - CFF
  - Statement of changes in Owners' equity
- **Importance of**
  - FS notes (footnotes)
    - disclose the basis of preparation for FS *(e.g: accounting methods, assumptions,...)*
    - Additional items:
      - acquisitions or disposals
      - legal actions
      - employee benefit plans
      - contingencies and commitments
      - significant customers
      - sales to related parties
      - segments of firm
    - are audited
  - Supplementary schedules
    - not audited
    - operating income or sales by region or business segments
    - reserves for an oil and gas company
    - info about hedging activities and financial instruments
  - MD&A
    - assessment of financial performance and condition of a company from the perspective of its management
    - Publicly held companies in US
      - Results from operations, with trends in sales and expenses
      - Capital resources and liquidity, with trends in CF
      - General business overview
    - discuss accounting policies that require significant judgements by management
    - discuss significant effects of trends, events, uncertainties
    - liquidity and capital resource issues, transactions or events with liquidity implications
    - Discontinued operations, extraordinary items, unusual or infrequent events
    - Extensive disclosures in interim financial statements
    - disclosure of a segment's need for CF or its contribution to revenues or profit
- **Audits of FS**
  - = independent review of an entity's FS
  - objective: auditor's opinion on fairness and reliability of FS, "no material errors"
  - Standard auditor's opinion
    - 3 parts
      - Independent review though FS prepared by mgmt and are its responsibility
      - Reasonable assurance of no material errors (follow generally accepted auditing standards)
      - FS prepared in accordance with accepted accounting principles, reasonable accounting principles and estimates, consistency
    - Explanatory paragraph: when a material loss is probable but amount cannot be reasonably estimated. Uncertainties may relate to the going concern assumption --> signal serious problems and need close examination by analyst
    - (under US GAAP): Opinion on internal controls
    - 3 types of Opinions
      - Unqualified opinion: auditor believes statements are free from material omissions and errors
      - Qualified opinion: if statements make any exceptions to accounting principles --> explain these exceptions
      - Adverse opinion: if statements are not presented fairly or are materially nonconforming with accounting standards
- **Other info sources**
  - Interim reports
    - Quarterly or semi- reports *(NOT audited)*
  - Proxy statements
    - About election of board members, compensation, management and qualifications and issuance of stock options
    - Filed with SEC
  - Corporate reports and press releases
    - Viewed as PR or sales materials
- **FSA framework**
  - 1. Articulate the Purpose & Context of analysis
  - 2. Collect data
  - 3. Process data
  - 4. Analyze/interpret data
  - 5. Report the conclusions or recommendations
  - 6. Update the analysis

23. Financial reporting mechanics

- **Classification**
  - Operating activity: activities that are part of the day-to-day business function of an entity
  - Investing activity: activities associated with acquisition & disposal of long-term asset
  - Financing activity: activities related to obtaining or repaying capital from shareholders or creditors
- **Account & financial statement**
  - FS elements & accounts
    - Elements
      - Assets
      - Liabilities
      - Equity
      - Revenue
      - Expense
    - Accounts
      - Chart of accounts : set forth the actual accounts used in a company's accounting system
      - Contra account: offset or deducted from other accounts
  - Accounting equation
    - Assets
      - Liabilities
      - Owners' equity
        - Contributed capital
        - Retained earning
    - Expanding: A = L + Contributed capital + BGN Retained earnings + Rev - Exp - Dividend
- **Accruals & Valuation adjustment**
  - Accruals
    - Cash movement prior to Acct. recognition
      - Unearned (Deffered) revenue
      - Prepaid expense
    - Cash movement after Acct. recognition
      - Unbilled (Accrued) revenue *(when billing, Un.Rev decrease & Receivables increase)*
      - Accrued expense
  - Valuation adjustment: made to company's A or L so that account records current market value (not *Historical cost)*
- **Relationships among IS, BS and statement of CFs, and of owners' equity**
  - BS: show a company's financial position at a point in time
  - Changes in BS accounts during an accounting period are reflected in IS, statement of CFs and owners' equity
- **Accounting system**
  - Flow of information
    - 1. Journal entries & Adjusting entries *(record=time)*
    - 2. General ledger & T-accounts *(record=order)*
    - 3. Trial balance *(list account balances at a particular point in time)*
    - 4. Fin. statement
  - Debit & Credit
- **Using fin. statement in security analysis**
  - Analyst uses FS to judge the fin. health of the company
  - Analyst can use his understanding to detect misrepresentation

24. Financial Reporting Standards

- **Overview FRS**
  - Objective of FR: provide fin. info about the reporting entity
  - Importance of reporting standards in security analysis and valuation
- **Standard setting & Regulatory bodies**
  - Standard-setting bodies (establishing standards)
    - IASB (International Accounting Standards Board)
    - US FASB (Financial Accounting Standards Board)
  - Regulatory authorities (enforcing standards)
    - IOSCO (international): *not a regulatory, but its members regulate significant portion*
    - FSA (in UK)
    - SEC (in USA)
      - 1. Protect investors
      - 2. Ensure: market is fair, efficient, transparent
      - 3. Reduce systematic risk
- **c.**
  - Status of global convergence of accounting standards
  - Barriers to developing one universally accepted set of financial reporting standards
    - disagree
      - standard setting bodies
      - regulatory authorities
    - political pressures from business groups and others
- **IFRS framework**
  - Qualitative characteristics
    - Relevance
    - Faithful presentation *(complete, neutral, free from error)*
    - Enhancing
      - Understandability
      - Verifiability
      - Comparability *(consistent among firms and time periods)*
      - Timeliness
  - Constraints
    - Trade off across Enhancing characteristics *(reliability and relevance: timely)*
    - Cost
    - Non-quantifiable info: omitted
  - Elements of FS
    - Measurements
      - of Financial position: A, L, E
      - of performance: Income, Expense
    - Assumptions
      - Accrual basis
      - Going concern
    - Recognition principal
      - Cost can be reliable measured
      - Probably future economic benefit will flow to entity
    - Measurement bases
      - Historical cost : amount originally paid for the asset
      - Current cost : would have to pay today for the same asset
      - Realizable value: amount for which firm could sell the asset
      - Present value : discounted future cash flows
      - Fair value : 2 parties in an arm's length transaction would exchange the asset
- **General requirements for FS under IFRS**
  - Required financial statements
    - BS, IS, CFS, OE, Explanatory notes (inclu. accounting policies)
  - Principles for PREPARING
    - Fair presentation
    - Going concern basis
    - Accrual basis
    - Aggregation
    - No offsetting
    - Consistency
    - Materiality
    - Comparative information
    - Frequency of reporting
- **IFRS (by IASB) # US GAAP (by FASB)**
  - Purpose of framework
    - IASB requires mgmt to consider the framework if no explicit standard exists
  - Objectives of financial statements
    - IASB same objective
    - FASB different objectives for biz and non-biz
  - Assumptions
    - IASB emphasizes going concern
  - Qualitative characteristics
    - Primary characteristics
      - FASB: relevance, reliability
      - IASB: comparability, understandability also
  - Financial statement elements
    - Performance
      - IASB: income+expenses
      - FASB: Revenues, Expenses, Gains, Losses, comprehensive income
    - Asset definition
      - IASB: resource from which future economic benefit is expected
      - FASB: future economic benefit
    - "Probable"
      - IASB: define criteria for recognition
      - FASB: define assets and liabilities
    - Values of assets to be adjusted upward
      - IASB: allow
      - FASB: not allow
- **Effective FR**
  - Characteristics of a coherent financial reporting framework
    - Transparency
    - Comprehensiveness
    - Consistency
  - Barriers to creating a coherent financial reporting framework
    - Valuation
    - Standard setting
      - Principles-based
        - IFRS
        - relies on broad framework
      - Rules-based
        - FASB in the past
        - specific guidance how to classify trx
      - Objectives oriented
        - FASB moving now
        - blend the other two
    - Measurement


To be continued...
For MORE CFA® Mind Maps, please go to:
http://www.e-junkie.com/ecom/gb.php?cl=274078&c=ib&aff=283565

45. Market Organization & Structure

- **Main functions of financial system**
  - Allow entities to
    - Saving
    - Borrowing
    - Issuing equity
    - Risk management
    - Exchanging assets
    - Utilizing information
  - Determine the returns that equate D &S
    - Equilibrium interest rate
  - Allocate capital to most efficient uses
- **Classification: Assets & Market**
  - Financial A vs. Real A
    - F.A: securities, currencies...
    - R.A: commodities, real estate...
  - Public vs. Private securities
    - Public sec: trade on exchanges
    - Private sec: not trade on exchange
  - Debt vs. Equity vs. Derivative
    - Debt
    - Equity
    - Der contract: values depend on the values of other assets
  - Primary vs. Secondary market
    - Primary: for newly issued sec
    - Secondary: subsequents sales of sec
  - Money vs. Capital market
    - Money: for debt securities < 1y
    - Capital: for equity+debt securities> 1y
- **Asset classes**
  - Securities
    - Equity
      - Common stock
      - Preferred stock
      - Warrants
    - Pooled investment vehicles
      - Mutual funds
      - ETFs and ETNs — sometimes refer as Depositories
      - ABS
      - Hedge funds
    - Fixed income — Convertible debt=F.I+Equity
  - Currencies
  - Contracts
    - Forward, Futures, Swap, Option
    - Insurance
      - Credit default swap
  - Commodities
  - Real assets
- **Financial intermediaries**
  - Brokers,Dealers & Exchanges
    - Brokers
    - Block brokers — *help large trades*
    - Investment banks
    - Exchanges
    - Alternative trading systems (ATS)
    - Dealers — *earn profit fr. bid-ask spread*
  - Securitizers
  - Depository institutions
  - Insurance companies
  - Arbitrageurs — *refer who buy A in 1 market & resell in another market*
  - Clearinghouses & Custodians
    - Clearinghouses: intermediaries between buyers & sellers
    - Custodians
- **Positions**
  - Long vs. Short
    - Long =Buy
    - Short =Sell
  - Short sales — *borrow securities & sell*
  - Leveraged positions
    - *borrow funds to buy A*
    - $$\text{Margin call } P = P_0 * \frac{1-\text{Initial margin}}{1-\text{Maintenance margin}}$$
- **Objectives of market regulation**
  - Protect unsophisticated investors
  - Establish minimum standard of competency
  - Help investors evaluate performance
  - Prevent insider
  - Promote commom FR requirements
  - Require minimum level of capital
- **Characteristics of well-functioning fin. system**
  - Complete market *(Availability)*
  - Operational efficiency *(Low cost)*
  - Informational efficiency *(P reflects fundamental info)*
  - Allocational efficiency *(at the best efficiency)*
- **Classification of markets**
  - Distinguish
    - Call market
      - Trades occur at specific times
      - All bids+asks are declared, and then one negotiated price is set for the stock
      - used
        - in smaller markets
        - to set opening prices and prices after trading halts on major exchanges
    - Continuous market
      - Trade occur any time the market is open
      - Price is set by
        - auction process
        - dealer bid-ask quote
  - Distinguish
    - Quote-driven markets (trade with dealers)
    - Order-driven markets
      - Matching rules
        - 1. Price
        - 2. Display precedence
        - 3. Time precedence
    - Brokered markets
- **Primary vs. Secondary markets**
  - Primary market
    - IPO vs. Secondary issues
    - Public offerings vs. Private placements
  - Secondary market
    - Securities trade after initial offerings
    - Importance: provide Liquidity+Price info
- **Order**
  - Market vs. Limit order
    - M.O: execute at the best P
    - L.O
  - Validity
    - Good-til-cancelled
    - Immediate-or-cancel
    - Good-on-close
    - Good-on-open
    - Stop order
      - Stop-sell
      - Stop-buy

46. Security Market Indices

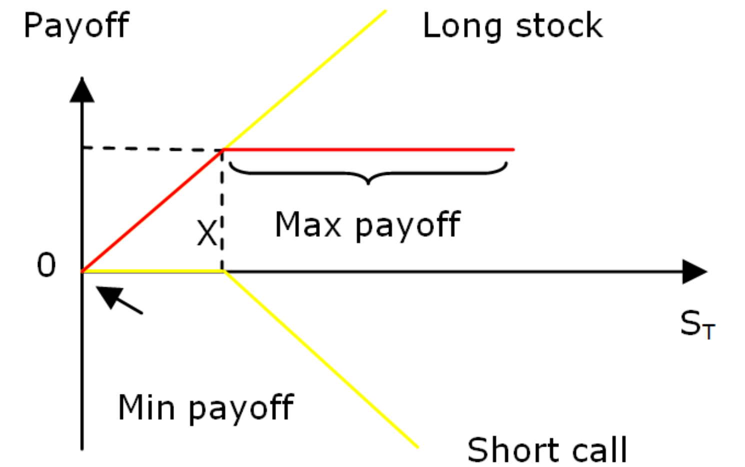
- **Security market index**
  - used to present the performance of an asset class, security market or segment of a market
- **Calculate an index**
  - Price index: calculate price only
  - Return index: include P+Income
- **Index construction & management**
  - Which target market?
  - Which securities?
  - How weight?
  - Re-balancing frequency?
  - Re-examining when?
- **Weighting methods**
  - Price-weighted index
    - = Sum of stock prices / Number of stocks adjusted for splits
    - Adv & Disad
      - Adv: simple
      - Disad: % change in a high-priced stock will have a greater effect on the index
    - Adjust for stock split
  - Equal-weighted index
    - Equivalent to a portfolio that has equal dollar amounts invested in each index stock
  - Market-cap weighted index
    - Weights based on the market-cap of each index stock
    - Criticism: large company has greater impact
    - Float-adjusted market cap- weighted index
    - Market float : (-) shares from Controlling shareholders
    - Free float: Market float - Not available to foreign investors
  - Equal-weighted index and Market-cap weighted index: NOT adjust
  - Fundamental weighting (earnings, dividends, cash flow)
- **Rebalancing & Reconstitution**
  - Rebalance: adjust the weights of securities
    - uses for Equal-weighted index
  - Reconstitution: add & delete securities that make up an index
- **Uses of securities market indices**
  - Reflect market sentiment
  - Proxy for measuring of market return & risk
  - Proxy of beta & risk-adjusted return
  - Benchmark of management performance
  - Model portfolio for index fund
- **Types of equity indices**
  - Broad market equity
  - Multi-market vs. Multi-market with fundamental weghting
  - Sector index
  - Style index
    - Market-cap
    - Value/Growth
- **Types of Fixed Income indices**
  - Large universe
  - Dealer market & infrequent trading
  - Illiquidity, transactions costs, high turnover of constituent securities => Difficult & expensive to replicate F.I index
- **Alternative investment indices**
  - Commodities index
    - *based on future contract*
  - Hedge fund index
    - *may have upward-bias*
  - Real estate index

57. Derivative Markets and Instruments

- **Definition**
  - Its return is based on another instrument (underlying assets)
  - Underlying assets
    - Physical
    - Finance
      - The biggest trading volume
    - Event
- **Where derivatives are traded?**
  - Exchange
    - Organized market -> liquid
    - Standard terms
    - No default risk
    - Daily settlement
  - OTC
    - private between 2 parties -> illiquid
    - Customized terms
    - default risk & legal risk
    - at the end of the contract: settlement
- **Characteristics**
  - Forward commitment
    - Firm and binding agreement -> obligation
    - No premium paid up front
  - Contingent claims
    - The long has the flexibility -> options
    - Premium is paid up front by the long
- **Types of derivatives**
  - Forwards
    - Exchange, OTC, Forward commitment
  - Futures
    - Exchange, Forward commitment
  - Options
    - Exchange, OTC, Contingent Claims
  - Swaps
    - OTC, Forward commitments
  - Credit derivatives
    - a contract that provides a bondholder (lender) with protection against a downgrade or a default by the borrower
    - Types
      - Credit default swap (CDS) -> most common
      - Credit spread option
- **Arbitrage & the law of one price**
  - Arbitrage
    - Buy an asset at one price
    - Concurrently sell it at higher price
    - -> Riskless profit without investment
  - The law of one price
    - NO arbitrage opportunities exist
- **Criticism**
  - Complex
    - Difficult to understand
  - Legal gambling
    - Zero-sum game
- **Purposes of derivatives market**
  - Price discovery
    - Information about underlying price
  - Risk management
    - Control risk
  - Market efficiency
    - Mispriced -> adjust quickly -> market efficiency
  - Trading efficiency
    - Low tnx cost

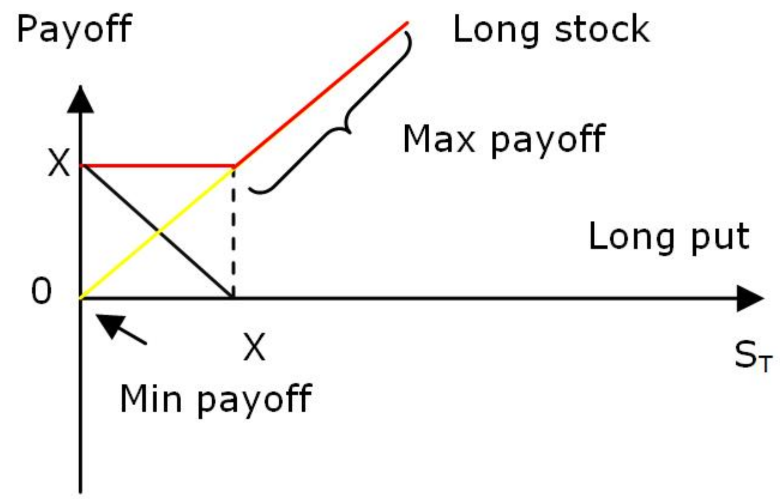
59. Risk Management Applications of Option Strategies

Covered call

- = Long stock + short call
  = S - C
- Covered call = call is covered by a long stock
- 

- Payoff diagram
- Payoff (covered call) = Payoff (Long stock) + Payoff (short call)
  $= S_T - \text{Max}(0, S_T - X)$
- Profit (Covered call) = Payoff (Covered call) - So + C
- Max loss when payoff is min -> $S_T = 0$ -> Max loss = So - C
- Max profit when payoff is max -> ST > X
- Payoff diagram (Covered call): similar to payoff diagram of short put

Protective put

- = Long stock + Long put
  = S + P
- Protective put = Long put protects potential loss of a stock
- 

- Payoff diagram
- Payoff (Protective put) = payoff (Long stock) + Payoff (long put)
  $= S_T + \text{Max}(0, X - S_T)$
- Profit = Payoff - So - P
- Max loss when payoff is min -> $S_T = 0$ -> Max loss = So + P - X
- Max profit when payoff is max -> ST > X -> Max profit is indefinite
- Payoff diagram (protective put) is similar to that of long call